



THE KING'S SCHOOL

2009
Higher School Certificate
Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1-7
- All questions are of equal value

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Total marks – 84

Attempt Questions 1-7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Use the table of standard integrals to find $\int \frac{1}{\sqrt{x^2 - 12}} dx$ **1**
- (b) Find the acute angle between the lines with gradients -9 and $\frac{4}{5}$. **2**
- (c) (i) Show that $\frac{1}{1 + e^{-x}} = \frac{e^x}{1 + e^x}$ **1**
- (ii) Hence, or otherwise, evaluate $\int_0^1 \frac{1}{1 + e^{-x}} dx$ **2**
- (d) Find the remainder $R(x)$ when $P(x) = x^3 + x^2 + x$ is divided by $x^2 - 12$ **2**
- (e) (i) Find the domain of the function $y = \ln(2x - 1) - \ln(x + 1)$ **2**
- (ii) Hence, or otherwise, solve the inequality $\frac{2x - 1}{x + 1} > 0$ **2**

End of Question 1

- (a) (i) Use the substitution $x = \sin\theta$ to show that

$$J = \int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx = \int_0^{\frac{\pi}{6}} \sin^2\theta d\theta \quad 3$$

- (ii) Hence, or otherwise, evaluate J . 3

- (b) The cubic equation $9x^3 + Ax + 2 = 0$, where A is real, has two roots whose sum is 1 and a third root α .

- (i) Find the value of A . 2

- (ii) Find the product of the two roots whose sum is 1. 1

- (c) (i) Write $\sin 2\theta$ in terms of $t = \tan\theta$ 1

- (ii) Prove that $\operatorname{cosec} 2\theta + \cot 2\theta \equiv \cot\theta$ 2

End of Question 2

(a) Find $\int \frac{dx}{9 + 4x^2}$ 2

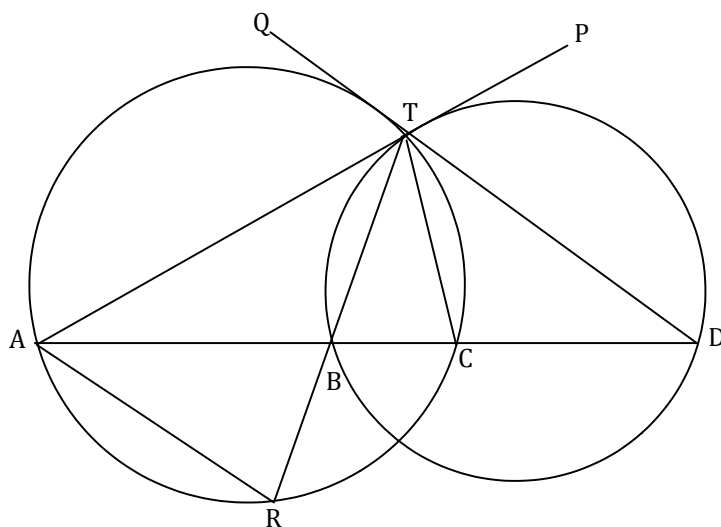
(b) The root of $f(x) = 0.4x - e^{-x^2} = 0$ is near $x = 1$
 Use Newton's Method once to find a 2 decimal place approximation to this root. 3

(c) A particle moves on the x axis so that at any time t its velocity is v .
 Prove that the acceleration $\frac{dv}{dt} = \frac{d\left(\frac{1}{2}v^2\right)}{dx}$ 2

(d) A particle is oscillating about $x = 0$ in simple harmonic motion.
 Its acceleration $\ddot{x} = -n^2x$, $n > 0$, and initially it is at rest at $x = A > 0$.
 (i) By using integration, prove that $v^2 = n^2(A^2 - x^2)$, where v is its velocity. 2
 (ii) When $x = 6$, $v = -2$ and when $x = 3$, $v = -4$.
 Find the amplitude and period of the motion. 3

End of Question 3

(a)



ATP is a tangent to the circle TBD.

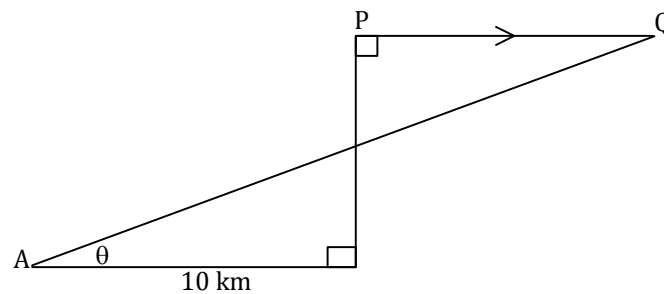
DTQ is a tangent to the circle TAC.

TB produced meets the circle TAC at R.

- | | |
|--|----------|
| (i) Explain why $\angle PTD = \angle TBD$. | 1 |
| (ii) Explain why $\angle PTD = \angle QTA$. | 1 |
| (iii) Deduce that $TB = TC$. | 2 |
| (iv) Prove that $\triangle ABR$ isosceles. | 2 |

Question 4 continues on next page

(b)



A plane P is flying due East at a constant height of 8 km and a constant speed of 6 km/min.

The plane is being tracked from a point A on the ground which was initially 10 km due West of the plane P.

Let Q be the position of the plane after t minutes and let the angle of elevation from A be θ at this time.

- (i) Show that $\tan\theta = \frac{4}{5 + 3t}$ 2
- (ii) Deduce that $\frac{d\theta}{dt} = \frac{-12}{\sec^2\theta(5 + 3t)^2}$ 2
- (iii) Find the rate at which the angle of elevation from A is decreasing after 1 minute. Give your answer in degrees/minute. 2

End of Question 4

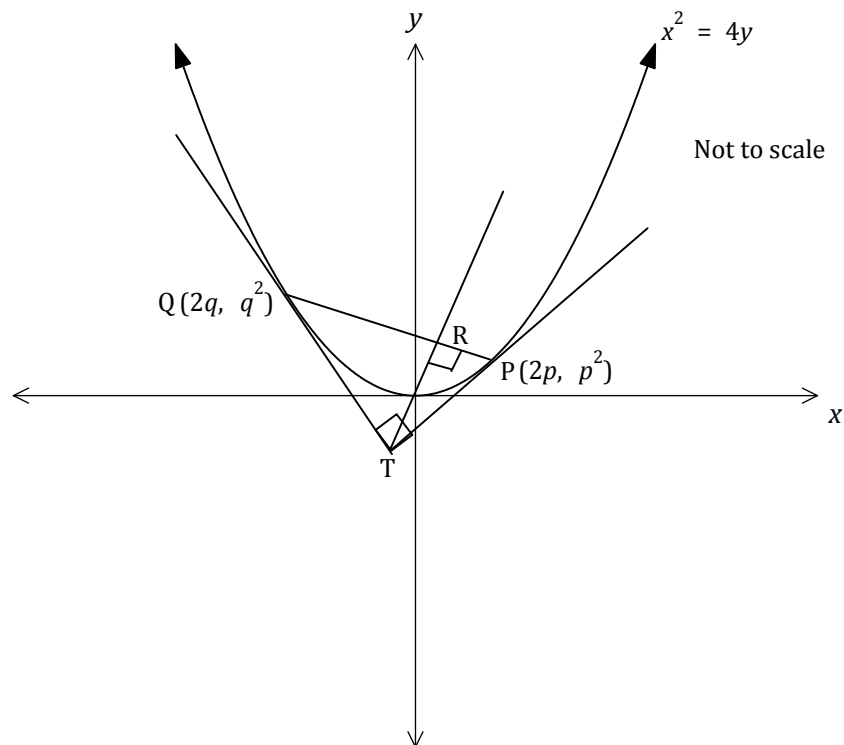
(a) (i) Show that $\frac{\binom{20}{k-1}}{\binom{20}{k}} = \frac{k}{21-k}$ **1**

- (ii) In the binomial expansion of $\left(x^2 + \frac{b}{x}\right)^{20}$ the coefficients of x^7 and x^4 are equal.

Find the value of b . **3**

Question 5 continues on the next page

(b)



The tangents at $P(2p, p^2)$ and $Q(2q, q^2)$ on the parabola $x^2 = 4y$ meet at right angles at T .

The equation of the tangent at P is $y = px - p^2$.

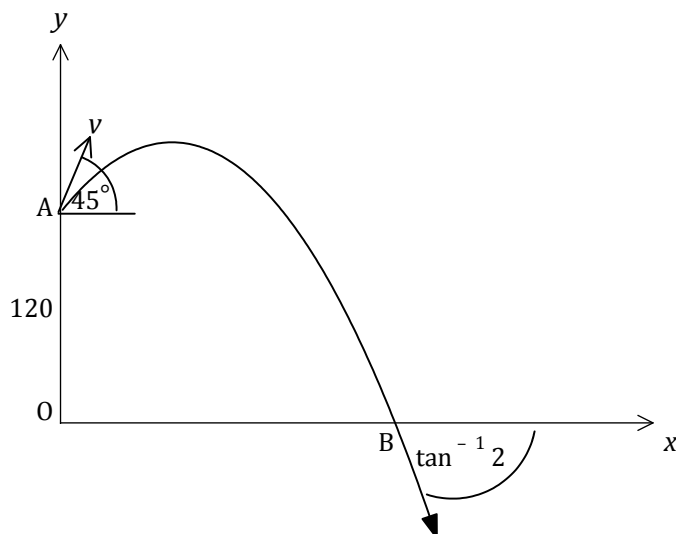
The equation of the chord PQ is $y = \frac{p+q}{2}x - pq$.

[DO NOT PROVE THESE]

- | | |
|--|----------|
| (i) Write down the gradient of the tangent at P . | 1 |
| (ii) Hence show that $pq = -1$. | 1 |
| (iii) Prove that the tangents at P and Q meet at $T(p + q, -1)$. | 2 |
| (iv) Show that PQ is a focal chord. | 1 |
| (v) A line is drawn from T to meet the chord PQ at right angles at R .
Prove that the equation of TR is $y = \frac{-2}{p+q}x + 1$. | 2 |
| (vi) Find the coordinates of R . | 1 |

End of Question 5

(a)



A projectile is fired from a point A, 120 m above horizontal ground, with a speed of v m/s and elevation 45° . It lands on the horizontal ground at B making an angle of $\tan^{-1} 2$ with the horizontal.

The equations of motion of the projectile are:

$$\begin{aligned} \ddot{x} &= 0 & \text{and} & & \ddot{y} &= -10 \\ \dot{x} &= \frac{v}{\sqrt{2}} & & & \dot{y} &= -10t + \frac{v}{\sqrt{2}} \\ x &= \frac{v}{\sqrt{2}} t & & & y &= -5t^2 + \frac{v}{\sqrt{2}} t + 120 \end{aligned}$$

[DO NOT PROVE THESE]

Suppose the projectile lands at B at time $t = T$.

- | | |
|--|---|
| (i) State the vertical component of velocity at B. | 1 |
| (ii) Prove that $10\sqrt{2} T = 3v$. | 2 |
| (iii) Find v . | 2 |
| (iv) Find the greatest height above horizontal ground that the particle reaches. | 2 |

Question 6 continues on the next page

(b) Let n be any integer ≥ 2 and x be any integer $\neq 0$

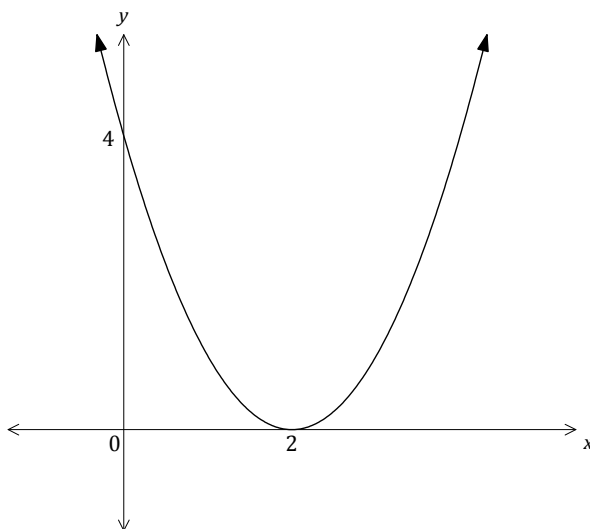
Let $E(n) = (1 + x)^n - nx - 1$

(i) Show that $E(n)$ is divisible by x^2 by using the binomial expansion. **1**

(ii) Prove that $E(n)$ is divisible by x^2 by using mathematical induction, $n \geq 2$ **4**

End of Question 6

(a)



The sketch shows the parabola $y = f(x) = (x - 2)^2$

(i) Explain why for $x \geq 2$ that an inverse function, $y = f^{-1}(x)$, exists. **1**

(ii) State the domain and range of this inverse function. **1**

(iii) At what point will $y = f(x)$ meet $y = f^{-1}(x)$? **1**

(iv) If $k < 2$, find, in simplest form, $f^{-1}(f(k))$ **2**

(b) Let $f(x) = 2\cos^{-1}x$, $-1 \leq x \leq 1$

and $g(x) = \sin^{-1}(2x^2 - 1)$, $-1 \leq x \leq 0$

(i) Sketch the graph of $y = f(x)$ **1**

(ii) Show that $g'(x) = f'(x)$, $-1 < x < 0$ **3**

(iii) Hence, or otherwise, express $g(x)$ in terms of $f(x)$. **2**

(iv) Sketch the graph of $y = g(x)$. **1**

End of Examination Paper

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$



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2009 Higher School Certificate Trial Examination

Mathematics Extension 1

Question	Algebra and Number	Geometry	Functions	Trigonometry	Differential Calculus	Integral Calculus	Total
1	(c)(i), (d), (e) 7			(b) 2		(a), (c)(ii) 3	12
2	(b) 3			(c) 3		(a) 6	12
3			(d)(ii) 3		(b), (c) 5	(a), (d)(i) 4	12
4		(a) 6	(b)(iii) 2	(b)(i) 2	(b)(ii) 2		12
5	(a) 4		(b) 8				12
6	(b) 5		(a) 7				12
7			(a), (b)(i), (iii), (iv) 9		(b)(ii) 3		12
Total	19	6	29	7	10	13	84

TKS EXTENSION 1 SOLUTIONS TRIAL 2009

Question 1

(a) $\ln(x + \sqrt{x^2 - 12}) \quad (+c)$

(b) $\tan \alpha = \frac{9 - \frac{4}{5}}{1 + 9 \cdot \frac{4}{5}} = \frac{45 - 4}{5 + 36} = 1$

$$\therefore \alpha = \frac{\pi}{4}$$

(c) (i) $\frac{1}{1 + e^{-x}} = \frac{e^x}{e^x(1 + e^{-x})} = \frac{e^x}{e^x + 1}$

(ii) $I = \int_0^1 \frac{e^x}{e^x + 1} dx = [\ln(e^x + 1)]_0^1$
 $= \ln(e + 1) - \ln 2$

(d) $x^2 - 12 \overline{) \begin{array}{r} x + 1 \\ x^3 + x^2 + x \\ \hline -12x - 12 \\ \hline 13x \end{array}} \Rightarrow Q(x) = x + 1$
 $\quad \quad \quad \frac{-12x}{13x} \quad \frac{-12}{12} \quad \vee \quad R(x) = 13x + 12$

(e) (i) $2x - 1 > 0$ and $x + 1 > 0$

ie. $x > \frac{1}{2}$ and $x > -1$

$\therefore$ domain is $x > \frac{1}{2}$

(ii) From (i), $y = \ln\left(\frac{2x-1}{x+1}\right)$

$$\Rightarrow \frac{2x-1}{x+1} > 0 \quad \text{if } x > \frac{1}{2} \quad \text{or } x < -1$$

Question 2

$$(a) \quad (i) \quad x = \sin \theta \quad x = 0, \theta = 0$$
$$\frac{dx}{d\theta} = \cos \theta \quad x = \frac{1}{2}, \theta = \frac{\pi}{6}$$

$$\therefore J = \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta \cos \theta}{\sqrt{1 - \sin^2 \theta}} d\theta$$
$$= \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta \cos \theta}{\sqrt{\cos^2 \theta}} d\theta = \int_0^{\frac{\pi}{6}} \sin^2 \theta d\theta$$

$$(ii) \quad J = \frac{1}{2} \int_0^{\frac{\pi}{6}} 1 - \cos 2\theta d\theta$$
$$= \frac{1}{2} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{6}}$$
$$= \frac{1}{2} \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} - (0) \right) \text{ will do}$$
$$= \frac{2\pi - 3\sqrt{3}}{24}$$

$$(b) \quad (i) \quad \text{Sum of roots} = 1 + d = 0 \quad \therefore d = -1$$

$$\therefore -9 - A + 2 = 0 \Rightarrow A = -7$$

$$(ii) \quad 2\beta\gamma = -\frac{2}{9} \Rightarrow \beta\gamma = \frac{2}{9}$$

$$(c) \quad (i) \quad \frac{2t}{1+t^2}$$

$$(ii) \quad \text{Put } t = \tan \theta$$

$$\text{Then } \operatorname{cosec} 2\theta + \cot 2\theta = \frac{1+t^2}{2t} + \frac{1-t^2}{2t}$$
$$= \frac{2}{2t} = \frac{1}{\tan \theta}$$
$$= \cot \theta$$

Question 3

$$(a) \int \frac{dx}{3^2 + (2x)^2} = \frac{1}{3} \tan^{-1}\left(\frac{2x}{3}\right) \cdot \frac{1}{2} \quad (+c)$$
$$= \frac{1}{6} \tan^{-1}\left(\frac{2x}{3}\right)$$

$$(b) f'(x) = 0.4 + 2xe^{-x^2}$$

$$\therefore x_1 = 1 - \frac{0.4 - e^{-1}}{0.4 + 2e^{-1}} = 0.97, \text{ 2 d.p.}$$

$$(c) \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} = v \frac{dv}{dx}$$
$$= \frac{d\left(\frac{1}{2}v^2\right)}{dv} \cdot \frac{dv}{dx} = \frac{d\left(\frac{1}{2}v^2\right)}{dx}$$

$$(d) (i) \frac{d\left(\frac{1}{2}v^2\right)}{dx} = -n^2 x$$

$$\therefore \frac{1}{2}v^2 = -n^2 \frac{x^2}{2} + c$$

$$\therefore 0 = -n^2 \frac{A^2}{2} + c, \quad c = n^2 \frac{A^2}{2}$$

$$\therefore v^2 = -n^2 x^2 + n^2 A^2 = n^2 (A^2 - x^2)$$

$$(ii) \therefore 4 = n^2 (A^2 - 36)$$

$$\text{and } 16 = n^2 (A^2 - 9)$$

$$\therefore \frac{A^2 - 9}{A^2 - 36} = 4 \Rightarrow A^2 - 9 = 4A^2 - 144$$

$$3A^2 = 135$$

$$A^2 = 45$$

$$\therefore \text{amp } A = \sqrt{45} = 3\sqrt{5}$$

$$n^2 = \frac{4}{45 - 36} = \frac{4}{9} \Rightarrow n = \frac{2}{3}$$

$$\therefore \text{period } T = \frac{2\pi \cdot 3}{2} = 3\pi$$

Question 4

(a) (i) Alternate segment theorem in circle BDT

(ii) Vertically opposite angles

(iii) $\angle QTA = \angle TCA$, alt. seg thm in circle TAC

$\therefore \angle TCA = \angle TBD$ from (i) & (ii)

$\therefore \triangle TBC$ is isosceles, base angles equal

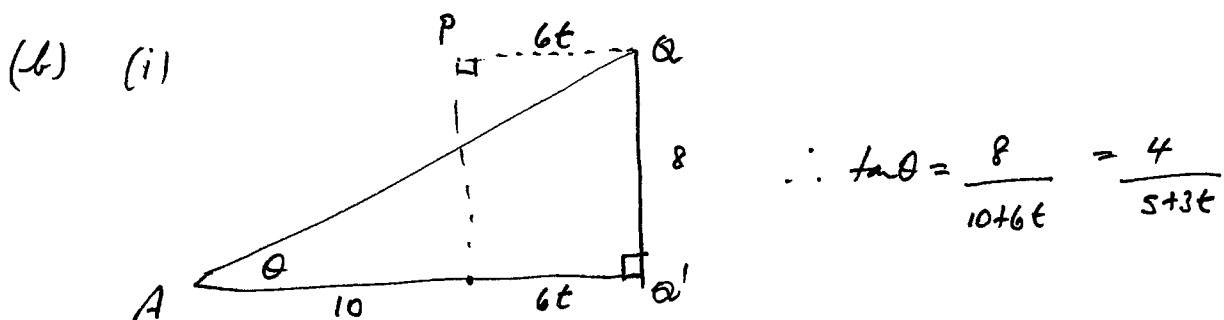
$\therefore TB = TC$

(iv) $\angle ABR = \angle TBC$, vert opp $\angle$ s

$\angle ARB = \angle TCB$, $\angle$ s in same segment, circle TAC

$\therefore$ From (iii), $\angle ABR = \angle ARB$

$\Rightarrow \triangle ABR$ is isosceles, base angles =



$$(ii) \frac{d \tan \theta}{dt} = \frac{d \tan \theta}{d \theta} \cdot \frac{d \theta}{dt} = \frac{d}{dt} 4(5+3t)^{-1}$$

$$\therefore \sec^2 \theta \frac{d \theta}{dt} = -4(5+3t)^{-2} \cdot 3$$

$$\text{i.e. } \frac{d \theta}{dt} = \frac{-12}{\sec^2 \theta (5+3t)^2}$$

$$(iii) t=1, \tan \theta = \frac{4}{8} = \frac{1}{2} \quad \therefore \frac{d \theta}{dt} = \frac{-12}{(1+\frac{1}{4}) 8^2} \text{ rad/min}$$
$$= -\frac{3}{20} \times \frac{180}{\pi} \text{ }^\circ/\text{min} = -\frac{27}{\pi} \text{ }^\circ/\text{min}$$

is decreasing at $\frac{27}{\pi} \text{ }^\circ/\text{min}$

Question 5

$$(a) \quad (i) \quad \frac{\binom{20}{k-1}}{\binom{20}{k}} = \frac{20! (20-k)! k!}{(21-k)! (k-1)! 20!} = \frac{k}{21-k}$$

$$(ii) \quad u_{k+1} = \binom{20}{k} (x^2)^{20-k} \left(\frac{b}{x}\right)^k \\ = \binom{20}{k} x^{40-3k} b^k$$

$$\text{Put } 40-3k=7 \Rightarrow k=11$$

$$40-3k=4 \Rightarrow k=12$$

$$\therefore \binom{20}{11} b'' = \binom{20}{12} b^{12}$$

$$\Rightarrow b = \frac{\binom{20}{11}}{\binom{20}{12}} = \frac{12}{21-12} \quad \text{from (i)} \\ = \frac{4}{3}$$

$$(b) \quad (i) \quad p$$

(ii) gradient of tangent at Q is q

$$\therefore pq = -1 \quad \text{since } \angle PTQ = 90^\circ$$

$$(iii) \quad \text{at } T, \quad px - p^2 = qx - q^2$$

$$\therefore (p-q)x = p^2 - q^2 = (p-q)(p+q)$$

$$\therefore x = p+q$$

$$y = p(p+q) - p^2 = pq = -1$$

$$\therefore T = (p+q, -1)$$

(iv) From data and (ii),

$$\text{chord } PQ \text{ is } y = \frac{p+q}{2} x + 1$$

$\Rightarrow (0,1)$ is on this chord

But the focus is $(0,1)$ $\therefore$ result.

(v) grad chord PQ is $\frac{p+q}{2}$

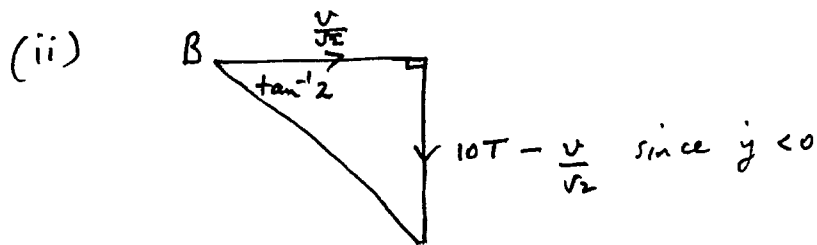
$$\therefore \text{gradient of } TR = -\frac{2}{p+q}$$

$\therefore$ TR is $y = -\frac{2}{p+q} x + 1$ since $(p+q, -1)$ is on it
and satisfies this equation

(vi) Comparing chord PQ and TR , clearly $R = (0,1)$, the focus.

Question 6

(a) (i) $\dot{y} = -10T + \frac{v}{\sqrt{2}}$



$$\Rightarrow 2 = \frac{10T - \frac{v}{\sqrt{2}}}{\frac{v}{\sqrt{2}}} = \frac{10\sqrt{2}T - v}{v}$$

$$\therefore 2v = 10\sqrt{2}T - v$$

$$\Rightarrow 10\sqrt{2}T = 3v$$

(iii) at B, $y = 0$

$$\therefore 0 = -5T^2 + \frac{v}{\sqrt{2}}T + 120$$

$$\Rightarrow 0 = -5 \cdot \left(\frac{3v}{10\sqrt{2}}\right)^2 + \frac{v}{\sqrt{2}} \cdot \frac{3v}{10\sqrt{2}} + 120 \quad \text{from (ii)}$$

$$\therefore -5 \cdot \frac{9v^2}{200} + \frac{3v^2}{20} + 120 = 0$$

$$\text{or } \frac{3v^2}{40} = 120 \quad \therefore v^2 = 40 \times 40$$

$$\therefore v = 40$$

(iv) $\dot{y} = 0 \Rightarrow t = \frac{v}{10\sqrt{2}} = \frac{4}{\sqrt{2}}$

$$\therefore \max y = -5 \cdot \frac{16}{2} + \frac{40}{\sqrt{2}} \cdot \frac{4}{\sqrt{2}} + 120 = 160$$

$$(b) (i) E(n) = (1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + x^n) - 1 - nx$$

$$= \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + x^n \quad \text{since } \binom{n}{1} = n$$

$$= x^2 \left(\binom{n}{2} + \binom{n}{3}x + \dots + x^{n-2} \right)$$

is divisible by x^2 (since $\binom{n}{2} + \dots + x^{n-2}$ is an integer)

$$(ii) E(2) = (1+x)^2 - 1 - 2x$$

$$= 1 + 2x + x^2 - 1 - 2x = x^2 \quad \text{is div. by } x^2$$

$\therefore$ Assume $(1+x)^n - 1 - nx = x^2 q$, q an integer, for $n \geq 2$

$$\text{Then } E(n+1) = (1+x)^{n+1} - 1 - (n+1)x$$

$$= (1+x)(1+x)^n - 1 - (n+1)x$$

$$= (1+x)[x^2 q + 1 + nx] - 1 - nx - x, \quad \text{using the assumption}$$

$$= q(1+x)x^2 + 1 + nx + x + nx^2 - 1 - nx - x$$

$$= q(1+x)x^2 + nx^2$$

$$= x^2 (q(1+x) + n)$$

is div by x^2 (since $q(1+x) + n$ is an integer)

$\therefore$ by induction $E(n)$ is div by x^2 for $n \geq 2$

Question 7

(a) (i) $\therefore f(x)$ is increasing for $x \geq 2$

(ii) $f(x) : x \geq 2, y \geq 0$

$\therefore f^{-1}(x) : x \geq 0, y \geq 2$

(iii) They must meet at $y = x$

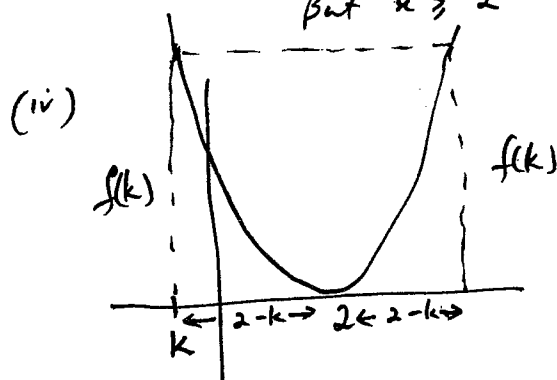
$$\therefore (x-2)^2 = x$$

$$x^2 - 5x + 4 = 0$$

$$(x-1)(x-4) = 0$$

$$x = 1, 4$$

But $x \geq 2 \therefore$ They meet at $(4, 4)$



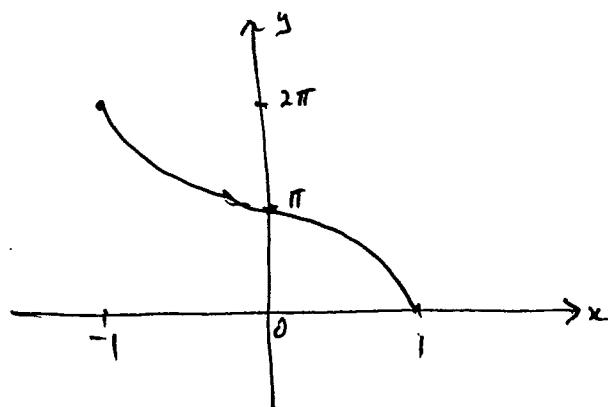
$$\therefore f^{-1}(f(k)) = 2 + 2 - k = 4 - k$$

[ALTERNATIVES ABOUND]

$$\text{e.g. } f(x) = (x-2)^2 = ((4-x)-2)^2$$

$$\Rightarrow \text{if } k < 2 \quad f^{-1}(f(k)) = f^{-1}(f(4-k)) \text{ where } 4-k > 2 \\ = 4-k$$

(b) (i)



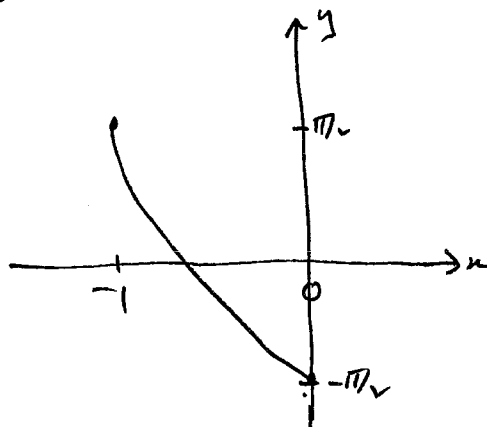
$$\begin{aligned} \text{(ii)} \quad g'(x) &= \frac{1}{\sqrt{1-(2x^2-1)^2}} \cdot 4x \\ &= \frac{4x}{\sqrt{1-(4x^4-4x^2+1)}} = \frac{4x}{\sqrt{4x^2-4x^4}} \\ &= \frac{4x}{2\sqrt{x^2}\sqrt{1-x^2}} \\ &= \frac{2x}{(-x)\sqrt{1-x^2}} \quad \text{since } -1 < x < 0 \\ &= \frac{-2}{\sqrt{1-x^2}} = f'(x) \end{aligned}$$

(iii) From (ii), $\sin^{-1}(2x^2-1) = 2\cos^{-1}x + C$, C a constant

$$\text{Put } x=0, \quad -\frac{\pi}{2} = \pi + C, \quad C = -\frac{3\pi}{2}$$

$$\therefore \sin^{-1}(2x^2-1) = 2\cos^{-1}x - \frac{3\pi}{2}, \quad -1 \leq x \leq 0$$

(iv) Using (i) and (iii),



Qn 1

(a) Ignore +c

(b) Allow 1 for $\tan x = \frac{9 - \frac{4}{5}}{1 + 9 \cdot \frac{4}{5}}$

(c) Allow 1 for $[\ln(e^x + 1)]'_0$

Allow 2 for decimal approxn [0.62 ---]

(d) Allow 2 for $x^2 - 12 \int \frac{x+1}{x^3 + x^2 + x} dx$
 $\frac{-12x}{13x+12} = R(x)$

(e) Allow 1 for $2x-1 > 0$ and $x+1 > 0$

Qn 2

(a) (i) Allow 1 for $\frac{dx}{d\theta} = \cos \theta$ & limits

Allow 1 for $J = \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta \cos \theta}{\sqrt{\cos^3 \theta}} d\theta$

(ii) Allow 1 for $J = \frac{1}{2} \int_0^{\frac{\pi}{6}} 1 - \cos 2\theta d\theta$

1 for $J = \frac{1}{2} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi/6}$

(b) (i) Allow 1 for $1 + x = 0$

(c) (ii) Allow 1 for $\frac{1+t^2}{2t} + \frac{1-t^2}{2t}$

(Alternatives exist, of course)

Qn 3

(a) Allow 1 for $\frac{1}{3} \tan^{-1}\left(\frac{2x}{3}\right)$

IGNORE +c

(b) Allow 1 for $f'(x) = 0.4 + 2xe^{-x^2}$

Allow 1 for $x_1 = \frac{1 - 0.4 - e^{-1}}{0.4 + 2e^{-1}}$ or equivalent

Allow 3 for $x_1 = 0.9717 \dots$

(c) Allow 1 for $\frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} = v \frac{dv}{dx}$

(d) (i) Allow 1 for $\frac{1}{2}v^2 = -n^2 \frac{x^2}{2} + c$ or equivalent
 i.e. $\frac{1}{2}v^2 = \left[-n^2 \frac{x^2}{2} \right]_A^x$

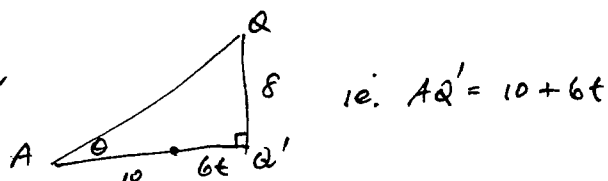
(ii) No marks for $4 = n^2(A^2 - 36)$ and $16 = n^2(A^2 - 9)$
 $\equiv$

Allow 1 for $\frac{A^2 - 9}{A^2 - 36} = 4$ or equivalent

Allow 2 for $A = \sqrt{45}$ and $n = \frac{2}{3}$

Qn 4

(b) (i) Allow 1 for



(ii) Allow 1 for $\frac{d \tan \theta}{d\theta} \cdot \frac{d\theta}{dt} = \frac{d}{dt} \left(\frac{4}{5+3t} \right)$ or clear indication of same

(iii) Allow 2 for $-\frac{27}{\pi}^\circ/\text{min}$ or decimal approx 8.594...

Qn 5

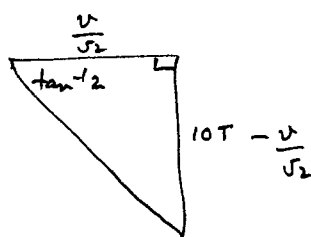
(a) (ii) Allow 1 for $u_{k+1} = \binom{20}{k} (x^2)^{20-k} \left(\frac{b}{x} \right)^k$

Allow 1 for $40-3k=7$ and $40-3k=4$

(b) (v) Allow 1 for reason why $\text{grad TR} = \frac{-2}{p+q}$

Qn 6

(a) (ii) Allow 1 for



(iii) Allow 1 for $0 = -5 \left(\frac{3v}{10\sqrt{2}} \right)^2 + \frac{v}{\sqrt{2}} \frac{3v}{10\sqrt{2}} + 120$ or equivalent

i.e. $-5T^2 + T \cdot \frac{10T}{3} + 120 = 0$

(iv) Allow 1 for $-10t + \frac{v}{\sqrt{2}} = 0$

IGNORE UNITS

(4) (ii) Allow 1 for $E(2)$

Allow 1 for clear assumption

IGNORE CONCLUSION

Qn 7 (a) (iv) No marks for $f^{-1}f(k) = k$

LOTS OF ALTERNATIVES, OF COURSE

e.g. $f^{-1}(x) : x = (y-2)^{\sqrt{}}$, $x \geq 0, y \geq 2$

$$\Rightarrow y = 2 + \sqrt{x} \quad [\text{DO NOT ALLOW 1 for this}]$$

$$\begin{aligned} \therefore f^{-1}f(k) &= f^{-1}((k-2)^{\sqrt{}}) = 2 + \sqrt{(k-2)^{\sqrt{}}} \\ &= 2 + 2-k \quad \text{since } k-2 < 0 \\ &= 4-k \end{aligned}$$

(4) (ii) Allow 1 for $g'(x) = \frac{1}{\sqrt{1-(2x^{\sqrt{}}-1)^{\sqrt{}}}} \cdot 4x$

$$1 \text{ for } g'(x) = \frac{4x}{2\sqrt{x^{\sqrt{}}} \sqrt{1-x^{\sqrt{}}}}$$

(iii) Allow 1 for $\sin^{-1}(2x^{\sqrt{}}-1) = 2\cos^{-1}x + c$